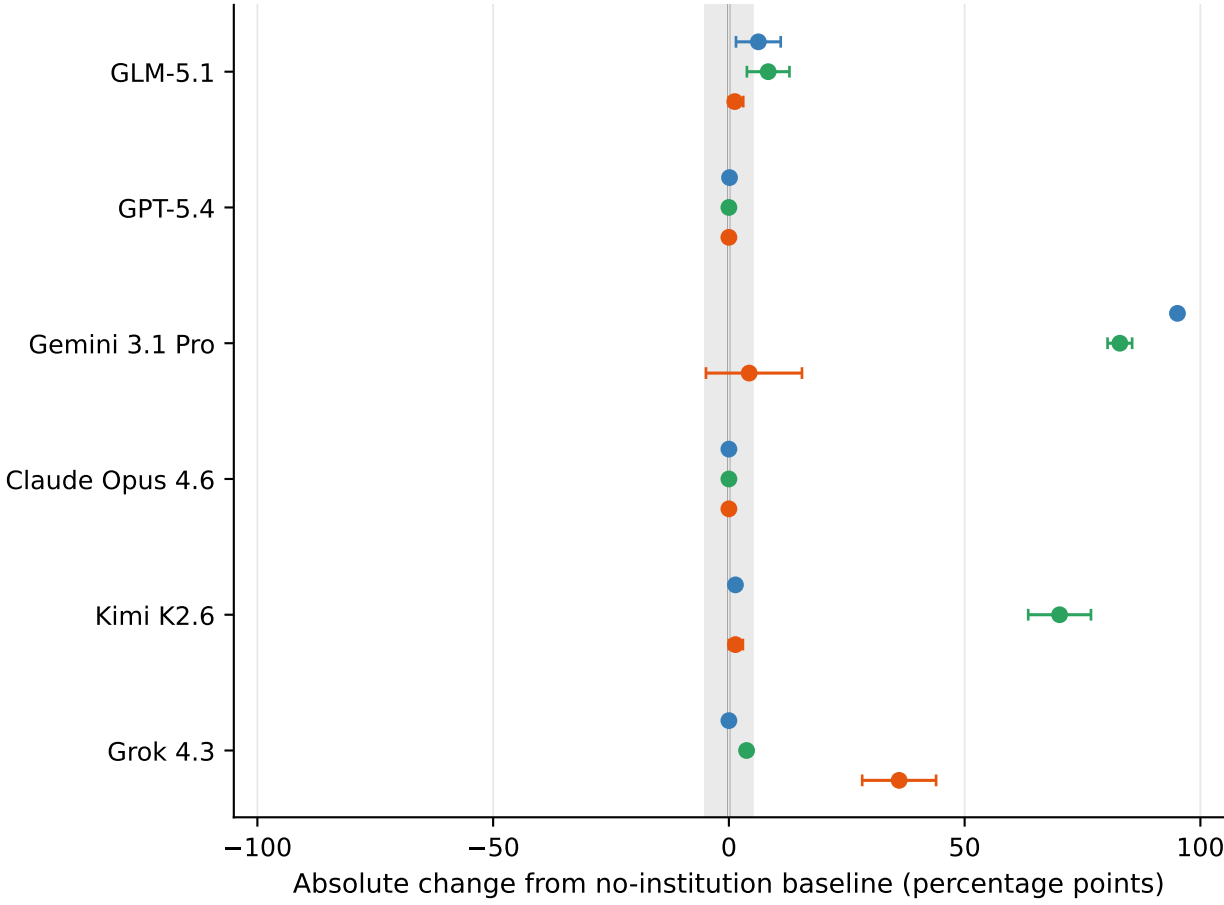


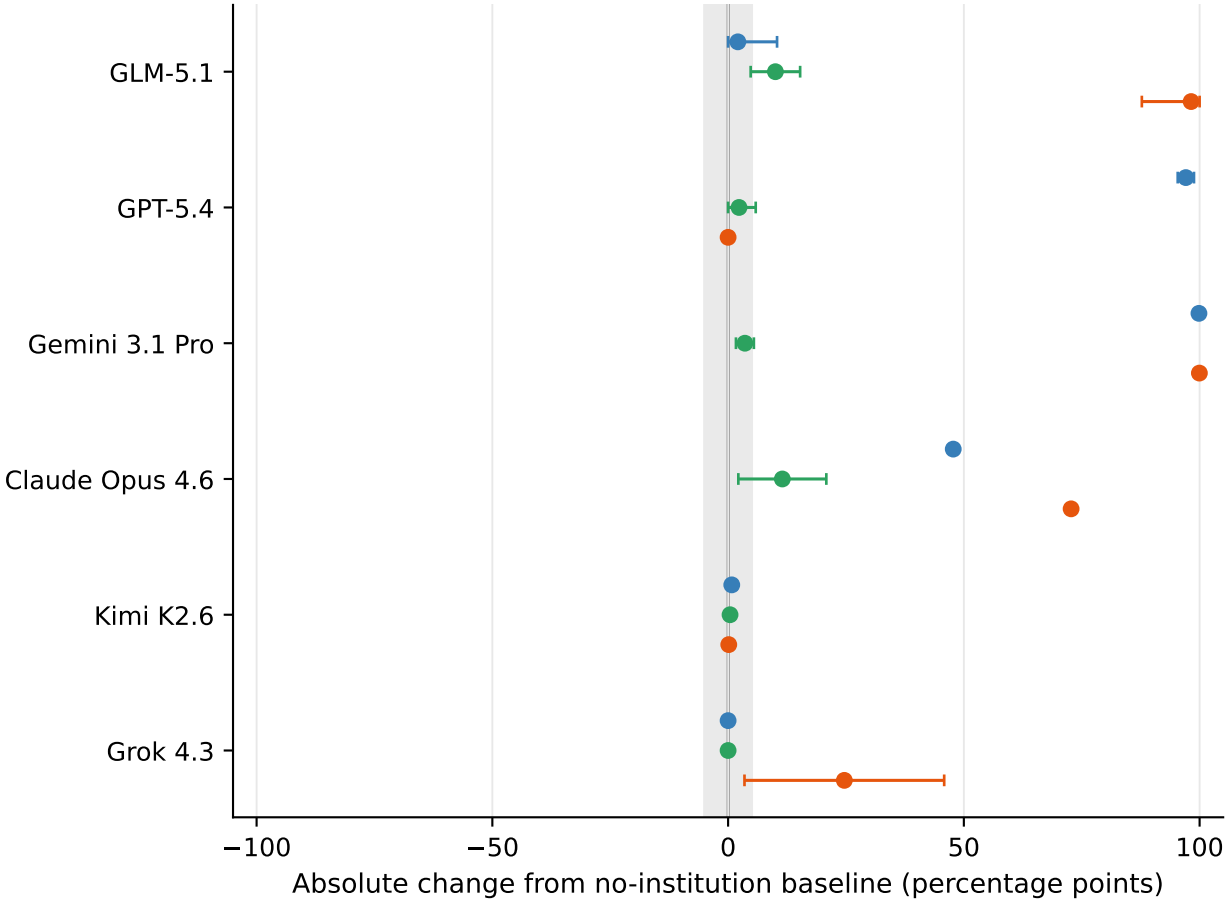
Figure 3. RQ2 institution effects on cooperation

Reputation Reward Punishment

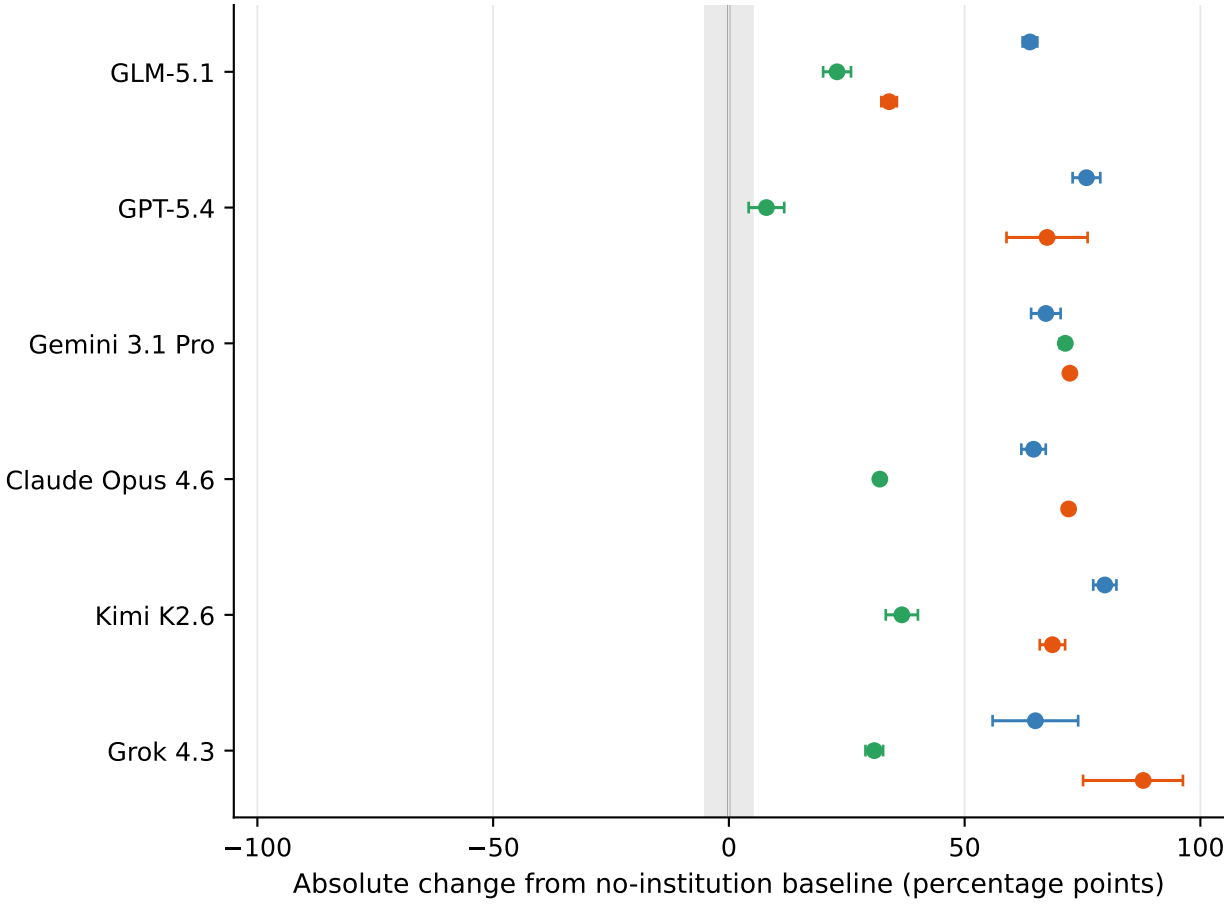
A Prisoner's dilemma: cooperation



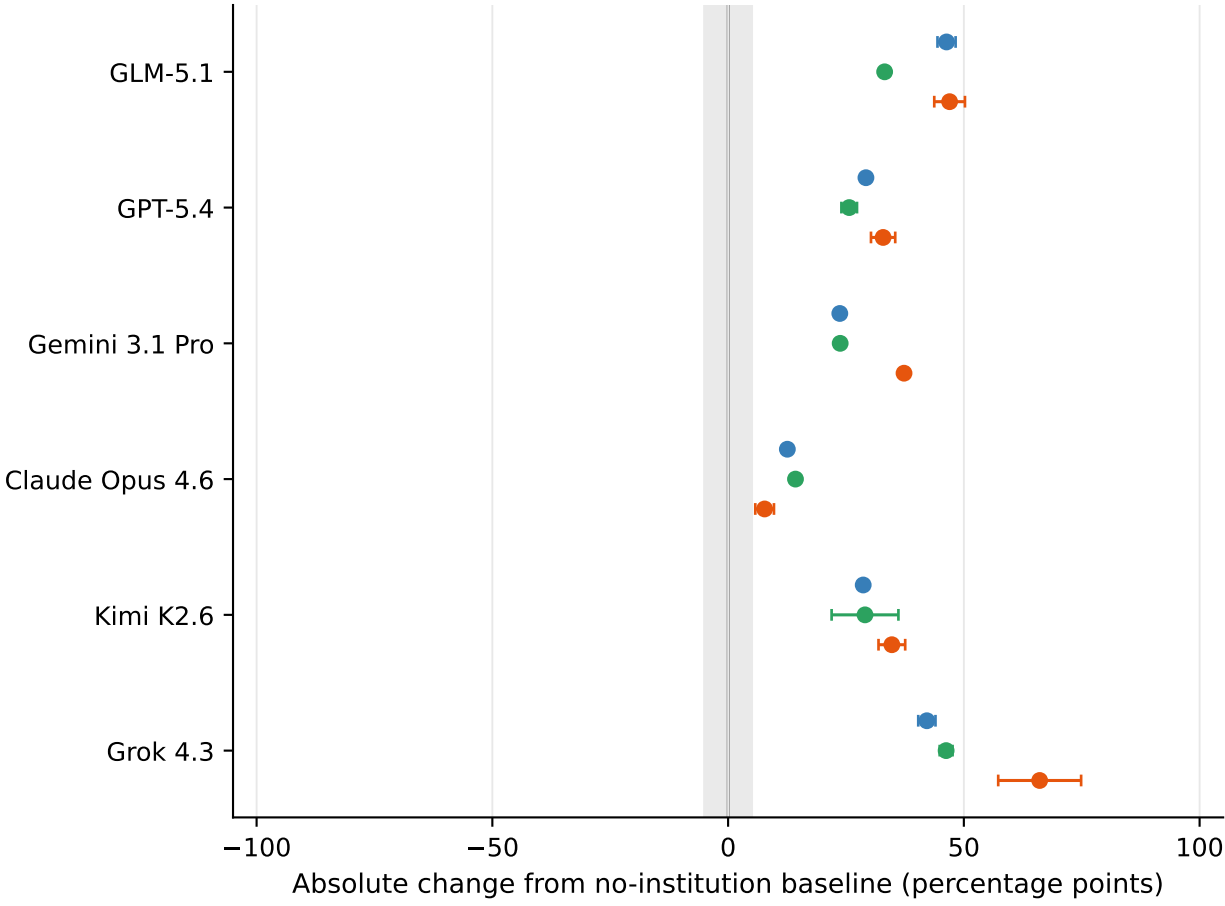
B Public-goods game: contribution



C Trust game: Sender transfer



D Trust game: Receiver conditional return



E Descriptive institution-effect types ($|\Delta| < 5$ pp = no material change)

	Reputation	Reward	Punishment
PD GLM-5.1	A	A	F
PD GPT-5.4	F	F	F
PD Gemini 3.1 Pro	R	R	N
PD Claude Opus 4.6	C	C	C
PD Kimi K2.6	F	R	F
PD Grok 4.3	F	F	A
PGG GLM-5.1	F	A	R
PGG GPT-5.4	R	F	F
PGG Gemini 3.1 Pro	R	F	R
PGG Claude Opus 4.6	R	A	R
PGG Kimi K2.6	F	F	F
PGG Grok 4.3	F	F	A
TG-S GLM-5.1	A	A	A
TG-S GPT-5.4	R	A	R
TG-S Gemini 3.1 Pro	A	A	A
TG-S Claude Opus 4.6	A	A	A
TG-S Kimi K2.6	R	A	R
TG-S Grok 4.3	R	A	R
TG-R GLM-5.1	R	A	R
TG-R GPT-5.4	A	A	A
TG-R Gemini 3.1 Pro	A	A	A
TG-R Claude Opus 4.6	A	A	A
TG-R Kimi K2.6	A	A	A
TG-R Grok 4.3	A	A	R

Institutional rescue Cooperation amplification No material effect Mechanism harm Ceiling Floor

Points are absolute changes from retained RQ1 no-institution means. Error bars are bounded 95% t intervals across treatment runs (baseline treated as fixed); gray band denotes ± 5 percentage points.